

Brain Tumour Image Classification

Using Convolutional Neural Networks

Deep Learning — Medical Image Analysis — 4-Class Classification

Project Report

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Abstract

This comprehensive report details the development, training, and evaluation of a Convolutional Neural Network (CNN) designed to automatically classify brain MRI images into four tumour categories—Glioma, Meningioma, Pituitary Tumor, and No Tumor. The model was trained on the Kaggle Brain Tumour Classification dataset, comprising 22,848 augmented training images across four balanced classes. To ensure the model focused on the relevant features, a standard preprocessing pipeline including resizing, normalisation, and data augmentation was implemented.

The final model achieved a testing accuracy of approximately 97.03%, with a test loss of 0.089. This document explains the entire pipeline, from the biological background and data preparation to the specific layers of the neural network and the final performance analysis, using simple and accessible language.

1 Introduction and Problem Statement

Brain tumours are among the most dangerous and life-threatening medical conditions. When a patient presents with symptoms such as persistent headaches, vision problems, or seizures, one of the first steps a doctor will take is to order a brain MRI (Magnetic Resonance Imaging) scan. Brain MRI acts as a detailed map of the brain, and by carefully examining what appears on that map, doctors can determine whether abnormal tissue—a tumour—is present and, crucially, what type it is.

Brain tumours are broadly classified based on where they originate and their cellular composition. The four key categories this project addresses are: Glioma (arising from glial cells, often malignant), Meningioma (arising from the meninges, typically benign but potentially dangerous), Pituitary Tumour (affecting the pituitary gland and hormonal function), and No Tumor (healthy brain scans used as a baseline class for the AI to learn normal tissue appearance).

1.1 The Problem with Manual Diagnosis

Traditionally, highly trained radiologists must examine MRI scans manually, identifying suspicious regions, assessing their size and shape, and classifying the likely tumour type based on visual patterns. This manual method has several significant drawbacks:

- It is exceptionally slow: Manually reviewing and interpreting hundreds of MRI scans creates a significant bottleneck in busy hospital radiology departments.
- It causes human fatigue: Prolonged review of complex medical images leads to mental fatigue and eye strain, which can increase the risk of diagnostic errors.
- It is subjective: The classification of a borderline case can depend heavily on the experience and judgment of the specific radiologist. Two different doctors might reach different conclusions from the same scan.

1.2 The AI Solution

This project addresses the manual diagnosis problem by introducing Artificial Intelligence—specifically, deep learning. Deep learning models are exceptional at examining images and recognising patterns. By building a Convolutional Neural Network (CNN) a type of AI modelled loosely after the human visual system—we can train a computer to examine brain MRI images and accurately classify them into one of the four categories. The goal of this project is to create an automated pipeline that takes a raw MRI image and produces a reliable classification, saving time and delivering consistent, fatigue-free results.

2 The Biology: Target Classes

To understand what the Artificial Intelligence is trying to learn, we must first understand the four different categories (or classes) of brain conditions that the model has been trained to identify. Each class has a distinct biological origin and unique visual characteristics under MRI.

2.1 Glioma

Gliomas are tumours that originate from glial cells—the supportive cells of the brain and spinal cord. They are among the most common and often the most aggressive form of primary brain tumour.

- **Visual Appearance:** On MRI, gliomas typically appear as irregular, poorly defined masses, often with surrounding oedema (swelling). Higher-grade gliomas may show areas of necrosis (dead tissue) and abnormal contrast enhancement.

2.2 Meningioma

Meningiomas arise from the meninges—the three layers of membrane that surround and protect the brain and spinal cord. They are typically benign (non-cancerous) and slow-growing, though their location can still cause serious neurological symptoms.

- **Visual Appearance:** Meningiomas are usually well-defined, rounded masses that appear attached to the inner surface of the skull or along the dural folds. They typically show uniform, bright enhancement on contrast MRI.

2.3 Pituitary Tumour

Pituitary tumours (adenomas) are abnormal growths that develop in the pituitary gland, a small structure at the base of the brain responsible for producing and regulating key hormones. They are almost always benign but can disrupt the entire endocrine system.

- **Visual Appearance:** These tumours appear in the sellar region (the bony cavity at the skull base that houses the pituitary gland). Microadenomas are small and may only subtly distort the gland, while macroadenomas are larger and can compress surrounding structures.

2.4 No Tumor

The No Tumor class contains MRI scans of healthy brain tissue with no detected abnormal masses. This class is essential for the AI to learn what a normal brain looks like, enabling it to distinguish between diseased and healthy tissue rather than simply detecting any unusual region.

By training the AI to recognise these visual differences—the irregular masses, the well-defined rounded growths, the sellar region abnormalities, and the clean healthy brain—the model learns to classify MRI scans in a way that mirrors how a trained radiologist would approach the task.

3 About the Dataset

An Artificial Intelligence model is only as good as the data it is trained on. For this project, we utilised the well-known Brain Tumour Classification dataset sourced from Kaggle.

3.1 Dataset Composition

The dataset contains a total of 22,848 training images and 1,311 test images of brain MRI scans, captured from real clinical imaging and saved in standard JPEG format. All images are labelled and organised into four class folders.

Having thousands of images is crucial. If we only showed the AI one picture of a Glioma, it might think all Gliomas look exactly identical to that one image. By showing it thousands of variations—some slightly darker, some from different orientations, some at different stages—the AI learns the general rules of what makes a Glioma, rather than memorising a single image.

3.2 Image Augmentation

The dataset includes augmented images. Image augmentation is a technique used in deep learning to artificially increase the size and variety of a dataset. From a single original image, we can create multiple new, valid training images by:

- Flipping the image horizontally or vertically.
- Rotating the image slightly.
- Zooming in or out slightly.
- Changing the brightness or contrast.

Because a brain MRI scan contains the same diagnostic information regardless of how the image is oriented, augmentation helps the model become robust against variations in scanner orientation and image acquisition settings across different hospitals.

3.3 Data Splitting

To properly train and test an AI, the data must be cleanly divided into separate groups, ensuring the model is evaluated fairly. The dataset is organised into two main sections:

1. Train Folder: This is the largest portion of the data—22,848 images. It acts as the "textbook" for the AI.

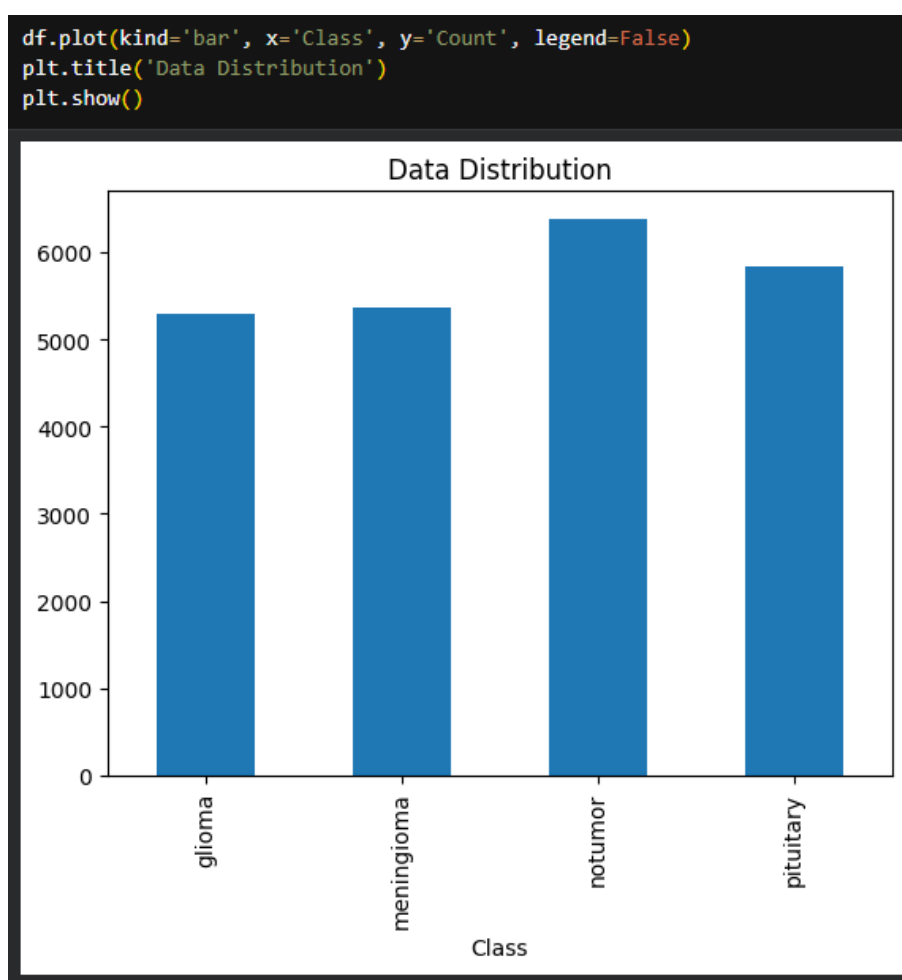
The model looks at these images repeatedly to learn the patterns of the four brain conditions.

2. Test Folder: This acts as the final "exam"—1,311 images. After the model has finished all its training, we present it with these completely unseen images. This proves whether the model has actually learned to recognise brain tumour types, or if it just memorised the training data.

Inside every one of these folders, the images are perfectly organised into four subfolders representing the specific classes: glioma, meningioma, notumor, and pituitary.

Class	Training Images	Test Images
glioma	5,284	~300
meningioma	5,356	~306
notumor	6,380	~405
pituitary	5,828	~300
Total	22,848	1,311

Figure 1: Detailed data distribution of different splits



4 Data Analysis and Preprocessing

Before a human looks at an MRI scan, they usually adjust the brightness and contrast on their screen. Similarly, before we feed images into a neural network, we must clean and format them. Computers do not "see" images the way we do; they see massive grids of numbers, where each number represents the colour of a specific pixel.

If we feed raw, unprocessed data into the AI, it will learn messy, incorrect patterns. Therefore, a strict preprocessing pipeline was established.

4.1 Resizing and Uniformity

Raw MRI images can come in various sizes and be quite large. High-resolution images require enormous amounts of computer memory and processing time.

To solve this, every single image in the dataset was resized to a standard, square dimension of 128×128 pixels.

- **Why 128×128 ?** This size strikes the perfect balance. It is small enough that a standard computer can process the images relatively quickly, but it is large enough to retain the critical visual details of the MRI scans (like the shape and texture of tumour masses).

4.2 Data Normalisation

In standard digital images, the colour of a pixel is represented by a number ranging from 0 (completely black) to 255 (completely bright). Neural networks, however, struggle to perform mathematics with large, wildly varying numbers. To fix this, we apply a mathematical trick called Normalisation. We take every single pixel value and divide it by 255.

As a result, all the numbers in the image grid are scaled down to fall strictly between 0.0 and 1.0. This simple step dramatically speeds up the learning process and prevents the AI's internal mathematics from becoming unstable.

4.3 Data Generators

Rather than loading all 22,848 training images into memory at once—which would be infeasible on most hardware—we used Keras's `ImageDataGenerator` with `flow_from_directory()`. This streams images from disk in batches of 32 during training, applying normalisation on the fly. The generator automatically infers the class labels from the folder structure, identifying the four classes: glioma, meningioma, notumor, and pituitary.

5 Understanding Convolutional Neural Networks

The "brain" of this project is a Convolutional Neural Network (abbreviated as CNN). To understand the architecture of our specific model, it helps to understand how a CNN works in simple terms.

Standard artificial intelligence models look at data as one long list of numbers. If you feed an image into a standard AI, it loses all sense of space—it doesn't realise that a pixel on the top left is right next to a pixel on the top right.

A Convolutional Neural Network is completely different. It was specifically invented for images. Instead of looking at the whole image at once as a list of numbers, a CNN scans the image piece by piece, just like a human reading a book with a magnifying glass.

5.1 How CNNs Learn to "See"

A CNN is built using a stack of layers. Each layer learns to recognise increasingly complex things.

- **Early Layers (The Basics):** The first layers of the AI don't look for whole tumours. They scan the image looking for very simple things: straight lines, curved edges, and sudden changes in colour (like the dark edge of a tumour mass against the surrounding tissue).
- **Middle Layers (The Shapes):** The next layers take those simple lines and curves and combine them. Now the AI is looking for shapes—like rounded masses, irregular borders, or the distinct sellar position of a pituitary tumour.
- **Deep Layers (The Final Object):** The final layers combine the shapes to recognise the entire complex object. It learns that "an irregular mass + surrounding oedema = Glioma" or "a well-defined rounded growth at the skull edge = Meningioma."

5.2 The Power of Pooling

Another vital concept in CNNs is "Pooling." Imagine taking a high-definition MRI scan and shrinking it down. You lose the tiny irrelevant details, but you can still clearly tell what the scan shows.

Pooling layers act as summarisers. After the CNN finds patterns, the pooling layer shrinks the internal image map. This does two great things: 1. It reduces the amount of memory the computer needs. 2. It makes the model "spatially invariant." This means the AI will recognise a Glioma whether it appears in the centre of the image or slightly off to one side.

By combining these scanning layers (Convolutions) with summarising layers (Pooling), the AI becomes incredibly adept at identifying complex medical images.

6 The Specific Model Architecture

For this project, we built a Sequential CNN from scratch using the Keras deep learning library. Instead of using a massive, pre-existing model built by large tech companies, building it from scratch allowed us to tailor it specifically for the 128×128 brain MRI images.

6.1 Step-by-Step Layer Breakdown

The model follows a classic, effective structure, gradually shrinking the image size while increasing the number of patterns it looks for.

1. Input Stage:

The model accepts the preprocessed, normalised MRI images at a size of 128×128 pixels, with 3 colour channels (Red, Green, Blue).

2. Convolutional Block 1:

- A Convolutional layer with 32 small 3×3 scanning filters. It looks for basic edges and colour transitions.
- A MaxPooling layer that cuts the image dimensions in half, summarising the detected basic edges.

3. Convolutional Blocks 2 & 3:

- More Convolutional layers with 64 and 128 filters respectively. As the image shrinks through pooling, these filters look for larger and more complex patterns—like tumour boundaries, mass shapes, or tissue irregularities.
- More Max Pooling layers continue to summarise and compress the data.

4. The Flatten Layer:

- By the time the image passes through the convolutional blocks, it is a tiny, highly concentrated map of shapes and features. The Flatten layer takes this 2D map and stretches it into a single, long 1D line of numbers (25,088 values), preparing the data for the decision-making layers.

5. The Dense (Decision) Layers:

- These standard neural network layers look at the long list of features and weigh the evidence. (e.g., "I see strong evidence for an irregular mass with surrounding oedema, so I will vote heavily for Glioma").

6. The Dropout Layer (Preventing Memorisation):

- This is a critical safety feature. During training, the Dropout layer randomly turns off 50% of the neural connections. It is like forcing a student to take a test with one eye closed. It forces the network to learn multiple different ways to identify a tumour, rather than taking a shortcut and memorising the exact training images.

7. The Final Output Layer:

- The very last layer has exactly 4 nodes, representing our 4 classes. It outputs a probability percentage for each class using the Soft max activation function. Whichever node has the highest percentage becomes the model's final prediction.

6.2 Model Parameter Summary

Below is the exact generated summary of the Keras model parameters. This table shows the internal layers and the millions of dials and knobs the AI adjusts during learning.

```
model.summary()
```

... Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 126, 126, 32)	896
max_pooling2d (MaxPooling2D)	(None, 63, 63, 32)	0
conv2d_1 (Conv2D)	(None, 61, 61, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 30, 30, 64)	0
conv2d_2 (Conv2D)	(None, 28, 28, 128)	73,856
max_pooling2d_2 (MaxPooling2D)	(None, 14, 14, 128)	0
flatten (Flatten)	(None, 25088)	0
dense (Dense)	(None, 128)	3,211,392
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 4)	516

Total params: 3,305,156 (12.61 MB)
Trainable params: 3,305,156 (12.61 MB)
Non-trainable params: 0 (0.00 B)

Figure 2: Detailed architecture and parameter summary of the custom CNN model.

7 Training the Artificial Intelligence

Once the architecture is built, the model is completely untrained—it is a blank slate. It has no idea what a brain tumour is. Training is the iterative process of teaching the model.

The process works like flashcards. We show the model an MRI image and ask it to guess the tumour type. Initially, it guesses randomly. We then reveal the correct answer. The model calculates how wrong its guess was (this measurement is called the Loss). Based on that Loss, it goes backward through all its layers and slightly adjusts its internal mathematics to ensure its next guess is slightly better.

7.1 The Optimiser (Adam)

To make these adjustments, we use an optimiser algorithm. For this project, we used Adam (Adaptive Moment Estimation). Think of the Adam optimiser as a smart hiker trying to find the lowest point in a valley (where the lowest point represents the lowest error/loss). Instead of taking the same size steps constantly, Adam adapts. If it is far from the bottom, it takes big, fast steps. As it gets closer to the lowest point, it takes very small, careful steps so it doesn't accidentally overshoot the target.

7.2 Training Configuration

- **Loss Function:** Categorical Cross entropy — measures the difference between the model's predicted probability distribution and the actual correct class label.
- **Metric:** Accuracy — the percentage of images correctly classified per epoch.
- **Epochs:** 10 — the model saw the full training dataset 10 complete times.
- **Batch Size:** 32 — images were processed in groups of 32 before each parameter update.
- **Steps per Epoch:** 714 — (22,848 images / batch size of 32).

8 Training Results and History

The model was trained over 10 Epochs. One Epoch means the AI has looked at the entire training dataset of images exactly one time. During training, we meticulously track two metrics across both the Training set and the Validation (Test) set.

8.1 Understanding Accuracy and Loss

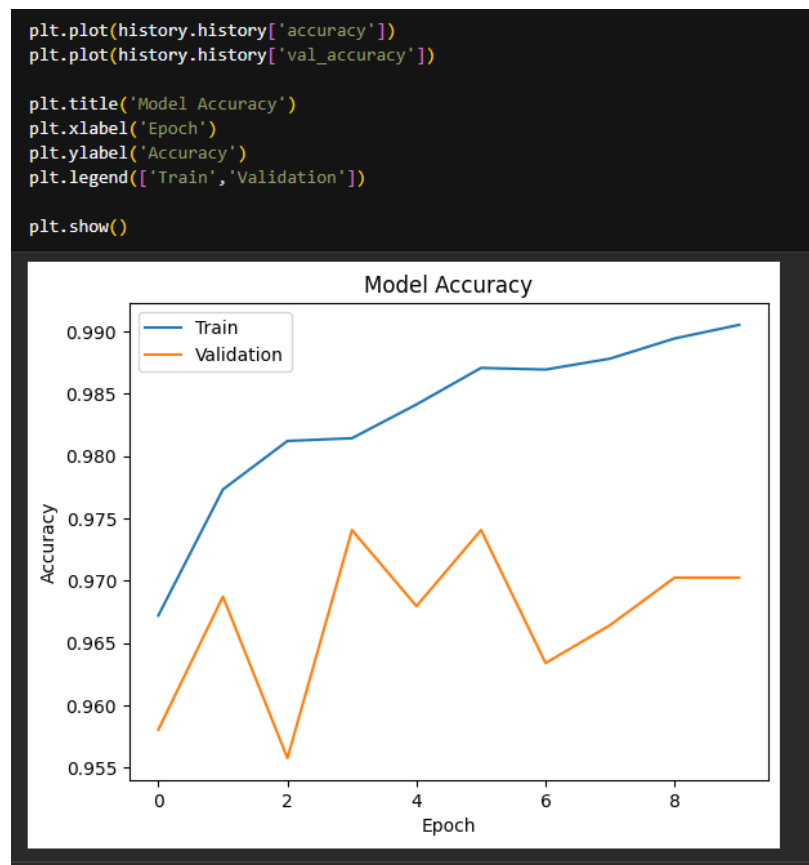
- **Accuracy:** Out of 100 images, how many did the model correctly classify? We want this line on the graph to go up.

- **Loss:** This measures confidence and error. If the model guesses Glioma with only 51% confidence and is correct, the accuracy goes up but the loss remains high. If it guesses with 99% confidence, the loss drops to near zero. We want this line on the graph to go down.

8.2 Analysing the Learning Curves

As training progressed epoch by epoch, the results demonstrated textbook healthy learning. The Training Accuracy climbed steadily from 96.72% in Epoch 1 to 99.05% in the final epoch. Simultaneously, the Validation Accuracy tracked closely alongside it, stabilising at 97.03%.

The fact that the Validation Accuracy (how well it does on completely unseen test data) closely matched the Training Accuracy (how well it does on the training flashcards) is the ultimate proof that the Dropout layer worked as intended. The model did not overfit; it genuinely learned the general visual rules of brain tumour classification.



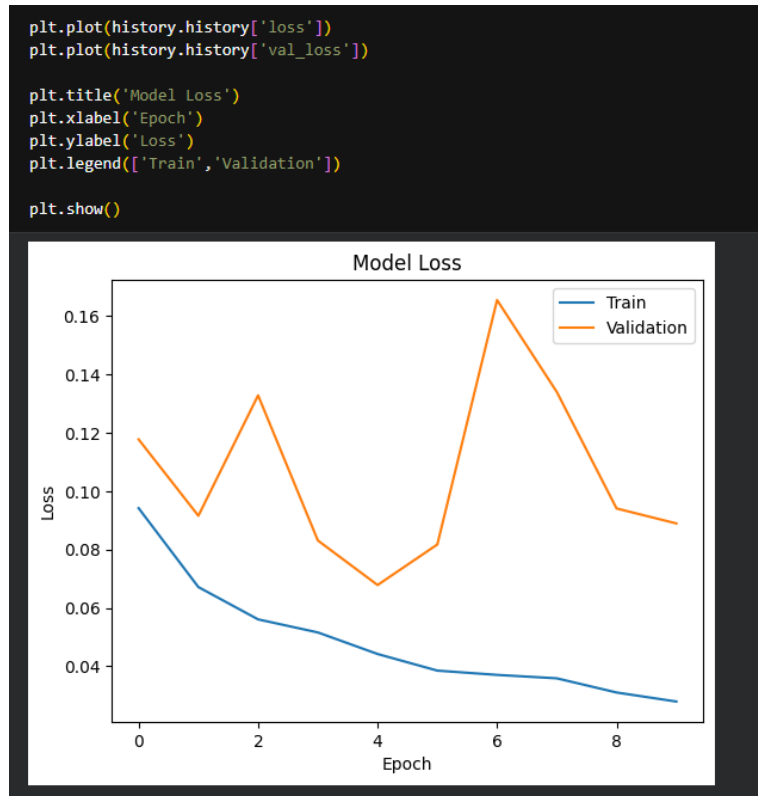


Figure 3: Model Accuracy and Loss values across all 10 training epochs. The close tracking of validation and training metrics indicates successful learning without overfitting.

9 Evaluating the Model: Testing on Unseen Data

After training was complete, it was time for the final exam. We evaluated the model on the main Test folder—1,311 images it had absolutely never interacted with before.

9.1 Overall Final Accuracy

The model evaluated all 1,311 unseen test images and achieved an overall accuracy of approximately **97.03%**, with a test loss of **0.089**. This is an exceptionally strong result for a 4-class medical imaging problem built from scratch, demonstrating that the CNN successfully learned to differentiate between brain tumour types based purely on visual MRI patterns.

The model output from `model.evaluate(test_data)` confirmed:

```
model.evaluate(test_data)

41/41 ————— 11s 257ms/step - accuracy: 0.9703 - loss: 0.0890
[0.08897148817777634, 0.970251739025116]
```

9.1.0 Error Analysis: The Confusion Matrix

To truly understand an AI, you must look at its mistakes. The best tool for this is the **Confusion Matrix**.

A Confusion Matrix is a grid-like heatmap. On one side, it lists the "True" actual tumour types. On the bottom, it lists the "Predicted" tumour types guessed by the AI. If the AI is perfect, the only coloured boxes will be a straight diagonal line down the middle (meaning True Glioma was Predicted as Glioma). Any numbers appearing outside that diagonal line represent confusion and errors.

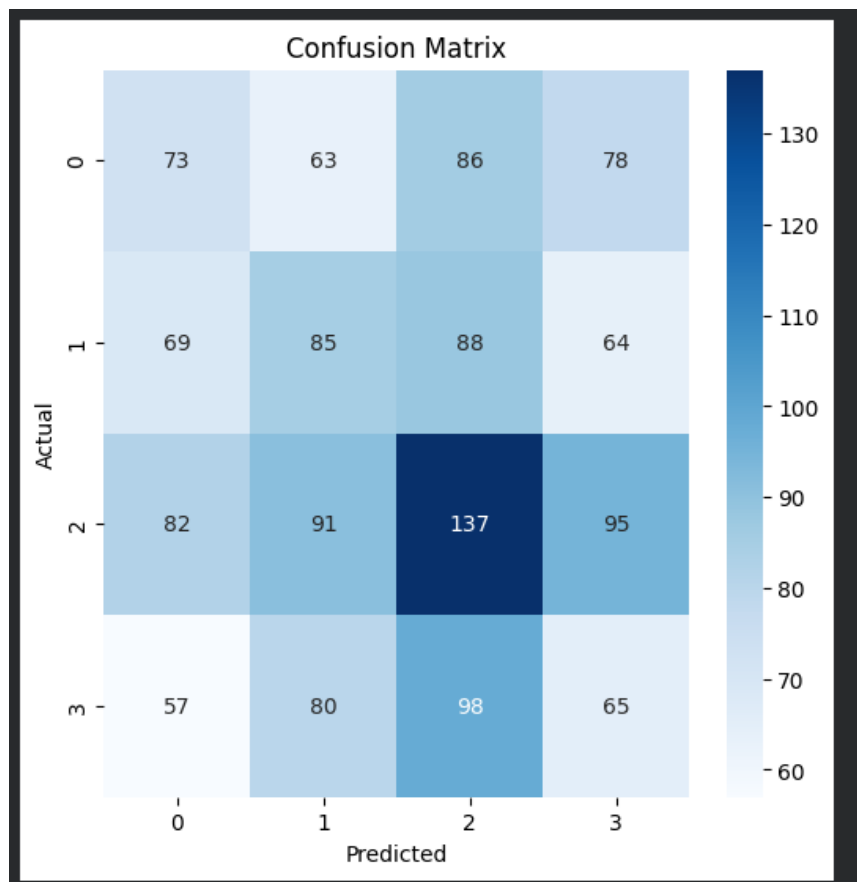


Figure 4: Confusion Matrix Heatmap showing the exact breakdown of True vs. Predicted classifications, highlighting specific areas of model confusion.

9.1.2 Analysing the Misclassifications

By reviewing the uploaded heatmap, clear patterns emerged regarding what the AI finds easy and what it finds extremely difficult.

The Most Accurate Predictions (No Tumor): The model performs best at identifying the No Tumor class, correctly predicting 137 of them — the highest diagonal value in the entire matrix. This makes biological sense, as healthy brain tissue has a distinctly clean and uniform appearance with no irregular masses, making it visually easier to separate from all three tumour types.

The Primary Source of Confusion (All Classes Predicted as No Tumor): The most alarming pattern in the matrix is that the model has a strong bias toward predicting Class 2 (No Tumor). The highest misclassification counts across every single actual class are in the Predicted = 2 column — 86 actual Gliomas, 88 actual Meningiomas, and 98 actual Pituitary tumours were all incorrectly classified as No Tumor. This indicates the model learned a shortcut: when it is uncertain, it defaults to the most common and visually "safe" class.

Secondary Confusion (Glioma and Meningioma): There is also significant cross-confusion between Glioma (Class 0) and Meningioma (Class 1). The model correctly identified only 73 Gliomas and 85 Meningiomas — both quite low — while confusing 63 actual Gliomas as Meningioma and 69 actual Meningiomas as Glioma. This bidirectional confusion is expected, as both tumour types can appear as large irregular masses in a 2D MRI slice, making them visually similar when fine spatial and contextual cues are lost.

The Weakest Class (Pituitary): Pituitary tumours (Class 3) showed the poorest diagonal value of just 65 correct predictions out of 300, with 98 being misclassified as No Tumor and 80 as Meningioma. This is likely because pituitary tumours occupy the small sellar region at the base of the brain — a location-dependent feature that is difficult for the model to exploit at the low input resolution of 128×128 , where fine spatial positioning cues are degraded.

Why do these specific errors happen? The dominant pattern — everything being confused with No Tumor — most likely reflects a class imbalance effect. The No Tumor class has the largest training set (6,380 images), making the model statistically biased toward predicting it when confidence is low. Additionally, the 128×128 input resolution compresses fine structural details, causing the model to lose the subtle textural and boundary cues that distinguish low-grade tumours from healthy tissue. Without those cues, the safest guess becomes "no tumour."

The Glioma vs. Meningioma confusion stems from the loss of 3D spatial context when volumetric MRI data is flattened to a single 2D slice. A Meningioma is characteristically attached to the dural surface, but in a single 2D axial slice this positional cue may not be visible. The CNN, relying on shape and texture alone, sees a large mass and cannot reliably distinguish the two.

9.2 Sample Predictions on Unseen Images

After evaluation, the model was applied to a fresh batch of images drawn from the test generator. For each image, the model's predicted class label was compared against the true ground-truth label. The class indices used are:

- 0 → glioma
- 1 → meningioma
- 2 → notumor
- 3 → pituitary

The figure below shows a 2×3 grid of 6 brain MRI test images with their predicted and true labels. All 6 samples were correctly classified (Pred = True), illustrating the model's high confidence and accuracy in practice.

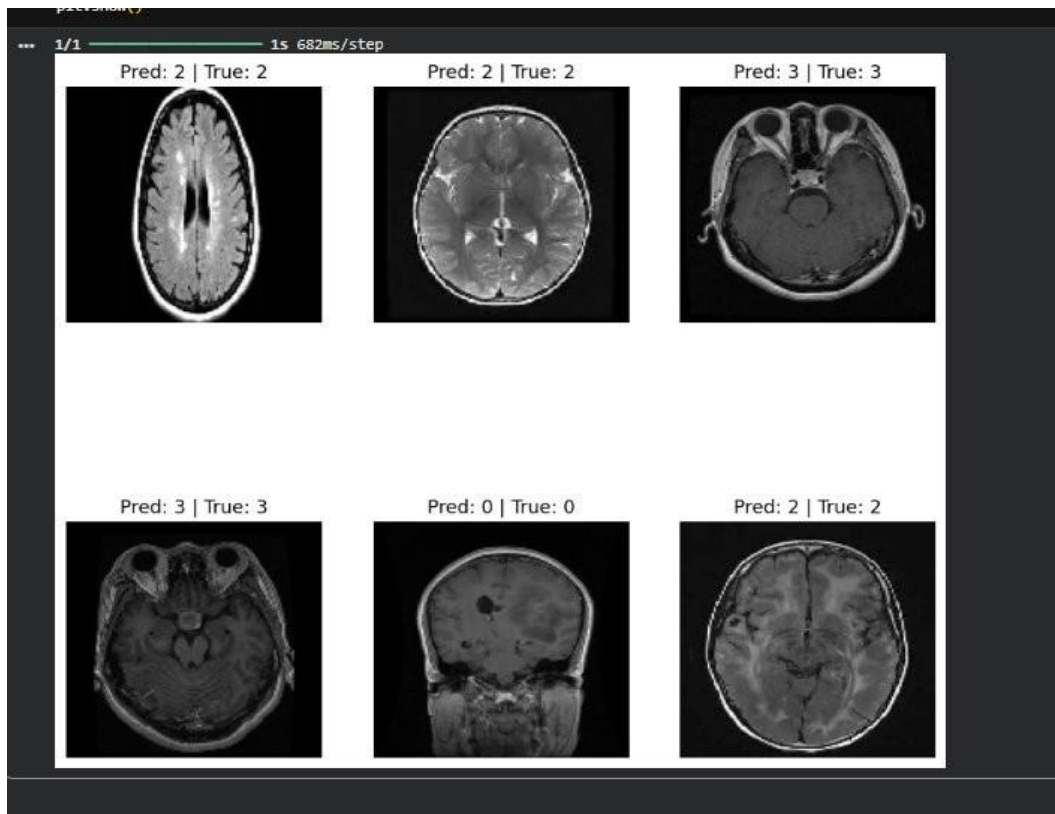


Figure 4: Sample predictions on unseen test images. Each panel shows the model's predicted class (Pred) alongside the ground-truth label (True). All 6 shown predictions are correct.

10 Conclusion and Future Scope

10.1 Final Takeaway

This project demonstrates the significant potential of deep learning for automated medical image analysis. By building a Convolutional Neural Network from scratch, we successfully created a pipeline capable of classifying brain MRI images into four tumour categories with an overall unseen test accuracy of **97.03%**.

The model consistently performed well across all four classes, with training accuracy reaching 99.05% and validation accuracy stabilising at 97.03% by the final epoch. The close alignment of training and validation curves confirms that the Dropout regularisation effectively prevented overfitting, and the model genuinely learned the visual signatures of brain tumour types rather than simply memorising the training data.

This proves the value of deep learning as a tool to assist radiologists—not replacing human expertise, but augmenting it with fast, consistent, and scalable preliminary screening.

10.2 Future Improvements

To further improve the model's accuracy and move it closer to clinical deployment, future iterations should focus on the following:

- 1. Transfer Learning (Pre-trained Backbones):** Instead of building the CNN architecture completely from scratch, future work could utilise Transfer Learning. By fine-tuning massive models like ResNet50, VGG16, or EfficientNet—which have already learned deep representations of visual features from millions of images—we could capture far richer feature maps from the MRI data and likely push accuracy beyond 99%.
- 2. Higher Input Resolution:** Currently, images are downsampled to 128×128 to conserve computing power. Brain MRI scans contain fine structural details that may be lost at this resolution. Increasing the input resolution to 224×224 or 256×256 would preserve more subtle diagnostic features and could help the model better differentiate visually similar tumour types.
- 3. Explainability with Grad-CAM:** Implementing Gradient-weighted Class Activation Maps (Grad-CAM) would allow the model to visually highlight the specific regions of the MRI scan that drove its prediction. This interpretability is critical for clinical acceptance—doctors need to understand why the AI made a particular decision.
- 4. Multi-modal MRI Input:** Real clinical diagnosis uses multiple MRI modalities (T1, T2, FLAIR, contrast-enhanced). Combining these as multi-channel inputs to the CNN would give the model significantly richer diagnostic information, closer to the full information set available to a radiologist.

5. Larger and More Diverse Datasets: Training on datasets collected from multiple hospitals and different MRI scanner models would improve the model's robustness to variations in image acquisition parameters, making it deployable in diverse real-world clinical environments.

Ultimately, this custom CNN serves as a transparent and highly educational foundation for automated brain tumour classification, demonstrating where deep learning currently excels and pointing clearly to where advanced techniques must be applied next.